SAMPLE

Earth's Sky: Clouds and Sky Phenomena

Rainbows, Halos, and More

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Introduction

The sky is not empty between the stars. By day it is a theater of clouds, the great engines of water and heat that make weather, and the light's repertoire: halos around the sun, rainbows after rain, rays that pour through broken cloud, mirages that bend the horizon, the green flash that ends a tropical sunset. At dusk the belt of Venus rises, the Earth's own shadow climbs the east, and at night, on rare and lovely occasions, the upper atmosphere itself glows with its own strange clouds. This book is about that whole repertory: the clouds and the optical phenomena, the sky's visible machinery, how it works, how to see it, and how people came to understand it.

Clouds and sky phenomena deserve a book together because they are one subject wearing two costumes. The clouds are the weather's hands, the visible tops of a hidden engine of heat and moisture; the optical phenomena are the sky's way of revealing its physics, every halo a hexagonal ice crystal, every rainbow a sphere of water doing geometry with the light. The two meet in the observer's sky: the cloud spotter who names the genus at noon is the halo watcher at sunset, and the hobby that begins with telling cumulus from stratus ends with knowing the circumzenithal arc on sight. The book serves both apprenticeships: the cloud classification from Luke Howard's Latin names to the World Meteorological Organization's modern atlas, and the optics from the common rainbow to the rarest arcs, with the science beneath each.

The chapters follow the discovery story first, how the naming of clouds became one of science's small revolutions and the explanation of the sky's colors became one of its satisfying ones, then the physics of water and light, the observer's craft, the field guide to the ten genera and the specialty formations, the context of art and folklore and forecasting, the modern science of clouds and climate, the frontier's unsolved cases, and the small delights of a sky stranger and lovelier than the casual glance ever suspects. The sky's phenomena are free to all and visible from any window; the book's purpose is to make the reader's glance casual no longer.

Two privileges set this sky's viewing apart, and they deserve the threshold's mention. The first is the price: the daytime and twilight sky's whole repertoire, the castellatus and the halo, the mammatus and the glory, costs nothing but the looking, no telescope, no filter, no travel beyond the nearest window or field. The second is the company: the sky performs constantly, over every city and every ocean at once, and the watcher who learns the grammar soon finds that no two days print the same page, that the same cumulus field never assembles twice, and that the discipline of noticing turns an ordinary commute into a running commentary the heavens provide free. The volume is an invitation into that company, and the invitation is genuinely open, which is rarer among the sciences than it should be.

What the reader will need, the chapters will supply: the naming history in the first chapter, the physics of water and light in the middle chapters, the field guides in the twelfth and thirteenth, and the trivia, the challenges, and the open questions between them. What the reader must supply, no chapter can: the habit of the upward glance, the patience to watch a cloud street rebuild itself after the wind shears it, and the small daily decision to treat the atmosphere as a text rather than a backdrop. The volume's promise, stated at the outset, is that the decision compounds: a month of deliberate looking changes the sky from weather into a language, and the language, once learned, is spoken everywhere on Earth for the rest of the watcher's life.

Chapter 1: The Discovery of Clouds and Sky Phenomena

The discovery story begins with the problem: everyone saw clouds, and no one could say what they were. The ancient writers, from Aristotle's treatise on the subject onward, described the sky's vapors and colors carefully but had no framework, and the weather's wisdom accumulated as folklore, the red sky at night, the halo before the storm, the mare's tails and mackerel scales of the old proverbs, true knowledge worn as rhyme. The obstacle was not observation but taxonomy: clouds shift and merge and dissolve, and a science needs its objects to have names, so the eighteenth century's meteorologists, a growing profession with instruments but no vocabulary, could measure the pressure and temperature while the sky's shapes stayed in the realm of the sketchbook.

The breakthrough came from an amateur, as the best stories have it. Luke Howard, a London pharmacist whose evenings belonged to the sky, read a paper in 1802, published in 1803, titled On the Modification of Clouds, and in it he did the thing no one had done: he named the clouds in Latin, the era's language of science, and he named their forms as modifications of one substance, water vapor, rather than as separate species. His four names, cumulus the heap, stratus the layer, cirrus the curl, and nimbus the rain-bearer, with the compounds the system allowed, gave the sky a grammar, and the naming took, the way good naming does, instantly and forever; the

scientific world adopted Howard's terms within a decade, the artists learned them, the poets wrote about them, and every cloud name used since is either Howard's or built in his method.

The naming's cultural echo deserves its own lines, because Howard's Latin entered the century's art and literature with astonishing speed. The Romantics took the clouds as subjects rather than scenery: the sky-studies of the painters, Constable's foremost, were painted with Howard's vocabulary in the notebooks, the science's names and the art's eyes on the same sky; the poets read the classification as a revelation, Shelley's and Goethe's clouds carrying the new taxonomy's wonder, Goethe's late correspondence with Howard himself the movement's emblem; and the century's landscape painting never looked back, the cloud becoming the sky's character actor, the weather's psychology, named and therefore seeable. The echo's lesson for the subject's history is the one this book keeps: the naming of a thing is the beginning of its culture, and the clouds' culture began the week the pharmacist's paper was read.
The nineteenth century built the edifice on Howard's foundation. The first International Cloud Atlas, coordinated by the meteorologists of the 1890s and published in 1896, the year of its era's great cloud photography, standardized the classification into genera and species and gave the observer an illustrated dictionary; the weather services of the world's nations trained their observers in it, and the sky's reports became comparable across the oceans. The same century explained the light: the rainbow's geometry yielded to the optics of refraction and reflection, Descartes and Newton's century-old work completed and corrected; the blue sky itself surrendered to Rayleigh's scattering in the 1870s and the sunsets' red along with it; and the halo family, catalogued by the century's patient observers, was traced to the ice crystal's hexagon by the 1920s, the great arcs' geometry computed and matched. The sky's phenomena, one by one, stopped being marvels and became physics.

The atlas's first edition deserves its own paragraph, because 1896 is the classification's other great date. The International Meteorological Committee's cloud commission, the era's cooperative project, assembled the atlas as the first international photographic standard: its plates, the cloud portraits commissioned from observatories around the world, gave the observer the reference images the text alone could not, and the classification became visual, the trained eye matching the sky to the plate. The atlas's later editions carried the standard through the century, the codes of the weather reports and the observer's manual being its working life, and the tradition the 2017 edition completed: the atlas that began as engraved plates is now a web gallery, the same idea, the sky's standard made of pictures, at two centuries' technological distance.
The twentieth century brought the clouds inside the laboratory. The cloud chamber, invented by the elder Wilson in 1911 for particle physics, was the condensation process made visible; the cloud physics that matured mid-century solved the growth problem, how a twenty-micron droplet becomes a raindrop in minutes, with the collision and ice-crystal processes; and the seeding experiments, from Vincent Schaefer's dry ice in 1946 onward, made the weather's mechanics an intervention's subject. The green flash's own history deserves its lines, the optics' most stubborn case. The flash was reported for centuries and doubted for most of them: the polar explorers' accounts of the distorted sun's strange colors, Byrd's Antarctic observation of the flash's whole sequence, were the testimony the skeptics finally had to read; the explanation's settlement, the refraction's dispersion and the mirage's magnification working together, came with the era's optics, and the flash's photographic era, the decades of the patient few capturing the emerald sliver, proved the commonplace what the folklore had called the rare. The history's lesson is the optics' own: the sky's phenomena are claimed, doubted, explained, and then collected, in that order, and every entry in the light's catalog has passed the same stations.
The optics completed their catalog: the green flash explained, the mirage families classified, the glory connected to the wave optics of the droplet's surface, and the rare arcs, the Parry and the tapestry of the cold-sky displays, computed and then photographed in the polar stations. The century also globalized the observer: the amateur cloud societies, the photography communities, and the sky-spotting culture that this book serves.

The story's modern act is the atlas's second life. The World Meteorological Organization's new International Cloud Atlas, the 2017 online edition, was the classification's first full revision in decades, and it did what the community of watchers had long asked: it added the volutus, the roll cloud, and the asperitas, the dramatic wave-cloud that the Cloud Appreciation Society's members had proposed from their photographs; it recognized the human-made clouds, the contrail's genus as cirrus homogenitus, and the clouds of wildfire's updrafts; and it moved the atlas to the web, where the observer's photograph can be the classification's evidence. The discovery story ends as it began, with the sky's watchers naming what they see, and the modern sky, contrails and all, now has its names.